

Name	Aravinthaa S
Register No	230701032
Dept & Sec	CSE A

Ex No – 11A	FIFO Page Replacement
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Program :

```
#include <stdio.h>

#define MAX 50

int main() {
    int page[MAX], frame[MAX];
    int n, m, i, j, k, f = -1;
    int page_faults = 0, flag;
    printf("Enter the size of reference string: ");
    scanf("%d", &n);
    printf("Enter the reference string:\n");
    for(i = 0; i < n; i++) {
        printf("Enter [%2d]: ", i + 1);
        scanf("%d", &page[i]);
    }
    printf("Enter number of page frames: ");
    scanf("%d", &m);
    // Initialize frames with -1
    for(i = 0; i < m; i++) {
        frame[i] = -1;
    }
    printf("\nFIFO Page Replacement:\n");
    for(i = 0; i < n; i++) {
        flag = 0;
```

```

// Check if page is already in frame
for(j = 0; j < m; j++) {
    if(frame[j] == page[i]) {
        flag = 1;
        break;
    }
}
// If page not found (page fault)
if(flag == 0) {
    f = (f + 1) % m;
    frame[f] = page[i];
    page_faults++;
    printf("\n%d -> ", page[i]);
    for(k = 0; k < m; k++) {
        if(frame[k] != -1)
            printf("%d ", frame[k]);
        else
            printf("- ");
    }
} else {
    printf("\n%d -> No Page Fault", page[i]);
}
}
printf("\n\nTotal Page Faults: %d\n", page_faults);
return 0;
}

```

Output :

```
cse32@localhost:~  
Enter the size of reference string: 20  
Enter the reference string:  
Enter [ 1]: 7  
Enter [ 2]: 0  
Enter [ 3]: 1  
Enter [ 4]: 2  
Enter [ 5]: 0  
Enter [ 6]: 3  
Enter [ 7]: 0  
Enter [ 8]: 4  
Enter [ 9]: 2  
Enter [10]: 3  
Enter [11]: 0  
Enter [12]: 3  
Enter [13]: 2  
Enter [14]: 1  
Enter [15]: 2  
Enter [16]: 0  
Enter [17]: 1  
Enter [18]: 7  
Enter [19]: 0  
Enter [20]: 1  
Enter number of page frames: 3  
  
FIFO Page Replacement:  
  
7 -> 7 - -  
0 -> 7 0 -  
1 -> 7 0 1  
2 -> 2 0 1  
0 -> No Page Fault  
3 -> 2 3 1  
0 -> 2 3 0  
4 -> 4 3 0  
2 -> 4 2 0  
3 -> 4 2 3  
0 -> 0 2 3  
3 -> No Page Fault  
2 -> No Page Fault  
1 -> 0 1 3  
2 -> 0 1 2  
0 -> No Page Fault  
1 -> No Page Fault  
7 -> 7 1 2  
0 -> 7 0 2  
1 -> 7 0 1  
  
Total Page Faults: 15  
[cse32@localhost ~]$
```

Ex No – 11B	LRU
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Program :

```
#include <stdio.h>
```

```
int findLRU(int time[], int n) {
```

```
    int i, minimum = time[0], pos = 0;
```

```
    for (i = 1; i < n; ++i) {
```

```
        if (time[i] < minimum) {
```

```
            minimum = time[i];
```

```

        pos = i;
    }
}

return pos;
}

int main() {
    int no_of_frames, no_of_pages, frames[10], pages[30];
    int counter = 0, time[10];
    int flag1, flag2, i, j, pos, faults = 0;
    printf("Enter number of frames: ");
    scanf("%d", &no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d", &no_of_pages);
    printf("Enter reference string: ");
    for (i = 0; i < no_of_pages; ++i) {
        scanf("%d", &pages[i]);
    }
    for (i = 0; i < no_of_frames; ++i) {
        frames[i] = -1;
    }
    for (i = 0; i < no_of_pages; ++i) {
        flag1 = flag2 = 0;
        // Check if page is already in frame
        for (j = 0; j < no_of_frames; ++j) {
            if (frames[j] == pages[i]) {
                counter++;
                time[j] = counter;
                flag1 = flag2 = 1;
            }
        }
    }
}

```

```

        break;
    }
}

// Page not in frame - empty slot
if (flag1 == 0) {
    for (j = 0; j < no_of_frames; ++j) {
        if (frames[j] == -1) {
            counter++;
            faults++;
            frames[j] = pages[i];
            time[j] = counter;
            flag2 = 1;
            break;
        }
    }
}

// No empty slot - use LRU
if (flag2 == 0) {
    pos = findLRU(time, no_of_frames);
    counter++;
    faults++;
    frames[pos] = pages[i];
    time[pos] = counter;
}

printf("\n");
for (j = 0; j < no_of_frames; ++j) {
    printf("%d\t", frames[j]);
}

```

```

    }

    printf("\n\nTotal Page Faults = %d\n", faults);

    return 0;
}

```

Output :

```

Enter number of frames: 3
Enter number of pages: 6
Enter reference string: 5 7 5 6 7 3

5      -1      -1
5      7       -1
5      7       -1
5      7       6
5      7       6
3      7       6

Total Page Faults = 4
[cse32@localhost ~]$

```

Ex No – 11C	OPTIMAL
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Program :

```

#include <stdio.h>

int i, j, nof, nor, flag = 0, ref[50], frm[50], pf = 0, victim = -1;

int optcal[50], count = 0;

int optvictim(int index);

int main() {

    printf("\n\nOPTIMAL PAGE REPLACEMENT ALGORITHM");

    printf("\n-----");

    printf("\n\nEnter the number of frames: ");

    scanf("%d", &nof);

```

```

printf("Enter the number of reference string elements: ");

scanf("%d", &nor);

printf("Enter the reference string: ");

for (i = 0; i < nor; i++)

    scanf("%d", &ref[i]);

// Initialize frame and calculation arrays

for (i = 0; i < nof; i++) {

    frm[i] = -1;

    optcal[i] = 0;

}

printf("\nReference String:\n");

for (i = 0; i < nor; i++)

    printf("%4d", ref[i]);

printf("\n\nProcessing...\n");

for (i = 0; i < nor; i++) {

    flag = 0;

    printf("\nref no %2d ->\t", ref[i]);

    // Check if page already in frame

    for (j = 0; j < nof; j++) {

        if (frm[j] == ref[i]) {

            flag = 1;

            break;

        }

    }

    if (flag == 0) {

        count++;

        if (count <= nof)

            victim++;

```

```

    else

        victim = optvictim(i); // Find optimal victim

        frm[victim] = ref[i];

        pf++; // Page fault
    }

    // Display current frame state
    for (j = 0; j < nof; j++) {
        if (frm[j] != -1)
            printf("%4d", frm[j]);

        else
            printf("  -");
    }
}

printf("\n\nTotal Page Faults: %d\n", pf);

return 0;
}

int optvictim(int index) {
    int i, j, temp, notfound;

    for (i = 0; i < nof; i++) {
        notfound = 1;

        for (j = index + 1; j < nor; j++) {
            if (frm[i] == ref[j]) {
                notfound = 0;

                optcal[i] = j;

                break;
            }
        }
    }
}

```

```

        if (notfound == 1)
            return i;
    }

    // Find frame with farthest next use
    temp = optcal[0];
    int pos = 0;
    for (i = 1; i < nof; i++) {
        if (optcal[i] > temp) {
            temp = optcal[i];
            pos = i;
        }
    }

    return pos;
}

```

Output :

```

OPTIMAL PAGE REPLACEMENT ALGORITHM
-----
Enter the number of frames: 3
Enter the number of reference string elements: 6
Enter the reference string: 6 5 4 3 2 1

Reference String:
  6   5   4   3   2   1

Processing...

ref no  6 ->      6   -   -
ref no  5 ->      6   5   -
ref no  4 ->      6   5   4
ref no  3 ->      3   5   4
ref no  2 ->      2   5   4
ref no  1 ->      1   5   4

Total Page Faults: 6

```